ENGINEERING / OPERATIONS HANDBOOK

One project.
Two flight computers.

Commands, retained execution failures, remote counter reset, firmware uploads and the current rocket/CanSat architecture.

Teensy 4.1 · Rocket | ESP32-WROOM · CanSat | Shared GCS | 41 command families

01 / FIRST USE

Start with the right vehicle

Ground control

1. Open workswithoutcmd copy.html in Chrome/Edge.
2. Connect the USB LoRa receiver at115200 baud.
3. Choose Rocket/CanSat under Control. Both telemetry streams stay visible; commands address one board.
4. Click Telemetry ON and wait for the selected flight ACK.

Mission workflow

Keep stationary, calibrate, set UTC if needed, ARM and wait. Mission elapsed starts at detected launch, not boot. Rocket recording must be stopped before ARM; start recording after arming.

Manual servos are available at any phase. One request at a time. An ACK never proves physical deployment or recording.

REMOTE COUNTER RESET / SELECT ONE VEHICLE FIRST

```
TELEM OFF              → wait for ACK
RESETCOUNT CONFIRM     → wait for durable DONE
TELEM ON              → first new packet1, new epoch
```

RESETCOUNT preserves mission/logs/calibration; it is not flash erase. Number ranges are reserved before transmitting; ordinary reboot gaps prevent reuse.

02 / POLICY

PAD-only gates removed. Execution checks retained.

Available beyond PAD

Telemetry OFF, calibration requests, RTC setting, reboot, camera start after landing and confirmed mission reset no longer fail just because of flight phase. Active ARM is idempotent: it reports the active mission instead of resetting it. Manual servos already bypass phase/sensor/storage gates.

Failures are still honest

Address/key/checksum/argument validation, sensor freshness/stillness, busy transactions, hardware failures, storage integrity and compile-mode restrictions remain. A valid command cannot invent sensor samples or a committed flash write.

Operator consequences: CAL BARO/ACCEL can change reference/filter. RESETPAD CONFIRM explicitly clears flight detection and automatic-action state even airborne. TIME changes UTC; REBOOT interrupts software. Removing the phase gate does not remove the physical consequence.

03 / INSTALL & STORAGE

What needs reflashing?

COMPONENT	ACTION FOR THIS UPDATE	PROCEDURE
Rocket	**Reflash required**	Arduino IDE: complete finalfsw/ folder, open finalfsw.ino, select Teensy 4.1 + USB Serial and upload.

COMPONENT	ACTION FOR THIS UPDATE	PROCEDURE
CanSat	**Reflash required**	Complete cansatfswfinal/ folder, open cansatfswfinal.ino, select classic ESP32 Dev Module/WROOM + correct port and upload.
Common GCS	**Reload updated HTML**	Needed for remote RESETCOUNT and durable ACK/epoch handling. No board flashing.
Ground receiver	No new receiver change	Keep current inspacelora/lora_rx/lora_rx.ino. If old, upload current version for its actual ESP32 pinout.
External W25Q64JV	**Do not erase for update**	MCU firmware upload and external-chip erase differ. Existing128byte records/CRC/commit and RFL1/CFL1 isolation remain compatible.

Local checks use Teensy core1.62.0 and ESP32 core3.3.12. Install board packages/libraries on each PC rather than copying AppData. Libraries: RadioLib, SparkFun BMP581 Arduino Library, RTClib + Adafruit BusIO; CanSat MPU9250_WE1.2.17 (contains MPU6500_WE); Rocket Adafruit BNO08x/TinyGPSPlus (TinyGPS++) and bundled Servo/SD/SerialFlash. Object checks do not create a fully linked/uploaded hardware build.

Fresh PAD routine telemetry defaults OFF. RF: MAIN865.50MHz / BACKUP867.25MHz, BW125kHz, SF7, CR5, sync0x95,14dBm. Switching is manual and must coordinate both ends. Vehicle IDs isolate commands but cannot prevent RF collisions.

▶ **Flash recovery, logging and factory erase**

04 / COMMAND REFERENCE

Find your command

Use NAME arg in the existing GCS Serial Write box; it supplies vehicle/key/sequence/XOR. Slashes and vertical bars in this reference mean choose one alternative; do not type the whole family label. Receiver USB needs authenticated flight frames; @RADIO is receiver-local. Direct flight-board USB accepts plain commands at115200/Newline.

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
Shared `HELP`	List implemented flight commands.	Help/status/ping remain accessible during pending operations. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `PING`	Read command RSSI/SNR; USB_OK if radio unavailable.	ACK is not proof of physical hardware behavior. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
Shared `STATUS`	Read numeric state, TX, sensors, reference/calibration and sequence status.	Unknown STATUS subtype falls back to summary; it is not another supported query. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS LINK`	Read radio/RX/TX flags and build marker.	RADIO=1 means initialized, not confirmed RF delivery. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS IMU`	Read model, freshness and errors.	Rocket BNO08x; CanSat MPU6500-compatible library at0x69. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS SD`	Read availability/file/write/drop/slow-operation counters.	CanSat ESP32 has no SD; MISSING is expected. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS RADIO`	Read channel/frequency/SF/sync/pending switch.	Receiver and addressed vehicle must share a channel. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS COUNTER / COUNT`	Read packet/next/high/epoch/TX/store blockers.	COUNT is a console alias; command sequence and telemetry counter are separate. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS FLASH`	Read JEDEC, journal state and16 header bytes.	Read-only, does not erase/repair. Read errors remain possible. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS MISSION`	Read mission ID/time quality/elapsed/recovery/storage/arming.	Uncertain recovered time is NA, not a fabricated timestamp. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `STATUS INA`	Read voltage/shunt/current/power and validity.	Invalid values remain explicit. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
Shared `TELEM ON / TELEM OFF`	Enable/disable routine LoRa telemetry, including in flight.	OFF preserves RX/ACK, sensors, flight logic, logs and timing. Real pending operations may be busy. Existing launch transition can force ON. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `RATE <400..10000>`	Set telemetry period, milliseconds.	Unsigned decimal in range; sensor/flight-loop rate unchanged. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `CAL BARO / GYRO / ACCEL / ALL`	Start3s calibration; await DONE/FAIL. BARO sets pressure reference; GYRO biases; ACCEL vertical bias.	PAD gate removed. Fresh valid IMU/pressure, stillness, sufficient samples, reference ordering and storage application still required. It can reset reference/filter if deliberately used in flight. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `TIME <1700000000..4102444799>`	Set/verify UTC Unix seconds on RTC.	No active-flight phase lock; RTC must exist, recovery must finish, value and readback must be valid. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `RADIO MAIN / RADIO BACKUP`	Coordinated vehicle/receiver retune; await DONE.	MAIN865.50MHz/BACKUP867.25MHz. SF7/sync0x95 unchanged. Actual radio/busy/pending-switch failures remain. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `ARM / ARM CONFIRM`	Initialize unstarted mission. Already active returns MISSION_ALREADY_ACTIVE without resetting.	PAD/LANDED phase gate removed. Fresh stationary IMU, baro reference, known counter, healthy storage and no conflicting operation remain. Recovery hold needs CONFIRM; bench1 cannot ARM. Rocket recording must be stopped first. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `RESETPAD CONFIRM`	Commit new unarmed PAD mission; keep logs/calibration/counter history.	Phase/active-flight gate removed. CONFIRM, idle healthy journal, mission-ID room and no conflicting operation remain. Can reset flight detection/automatic-action state while airborne. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
Shared `REBOOT`	ACK then software reset after bounded storage drain.	PAD/unarmed gate removed. Recovery must finish; pending camera/calibration must finish. Unsynced SD rows can be lost after drain deadline. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `COUNTER <last_used_upper_bound>`	One-time legacy packet-number migration.	Only when baseline is missing. Nonzero unsigned floor must cover stored high bound and be below UINT32_MAX; does not reset a healthy managed counter. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Shared `RESETCOUNT CONFIRM`	Remote selected-vehicle counter reset with durable new epoch.	First TELEM OFF. Requires known counter, completed recovery, idle healthy journal/radio, no conflicting transaction. Wait for PERSISTING then DONE. Mission/logs/calibration unchanged. First packet1 after ON/new reservation. finalfsw/finalfsw.ino; cansatfswfinal/cansatfswfinal.ino
Rocket `CAM 1 / CAM 0`	Request recording start/stop via Serial5.	Both send experimental CC0101E7 once; not verified recording-specific camera feedback. Camera transaction/storage/UART failures remain. Start after LANDED no longer requires a new mission; old landing timer will not immediately toggle a late manual start. finalfsw/finalfsw.ino
Rocket `CAMPWR`	Send camera power-button packet once.	No launched-only gate. Recording/camera-busy/recovery checks remain. No verified explicit power-off sequence. finalfsw/finalfsw.ino
Rocket `STATUS CAMERA`	Read request/UART/firmware-ACK/landing-stop flags.	Physical camera state unknown. finalfsw/finalfsw.ino
Rocket `PWM4 <0..270deg | 500..2500us>`	Rocket servo pin22.	Manual-anytime. Nominal500..2500us spans user-reported270deg MG996R. Automatic linkage unconfigured. finalfsw/finalfsw.ino

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
Rocket `PWM5 <0..270deg | 500..2500us>`	Rocket servo pin23.	Manual-anytime. Automatic stow0/deploy90 enabled at existing durable APOGEE-to-DESCENT event. finalfsw/finalfsw.ino
Rocket `PWM4/PWM5 TEST90 / SAFE`	TEST90=90deg; SAFE=configured target.	PWM4 SAFE90/1167us; PWM5 SAFE0/500us. SPIN/STOP/REVERSE no longer valid for positional servos. finalfsw/finalfsw.ino
Rocket `STATUS SERVO`	Read outputs, commanded pulses, automatic status/intent.	No position feedback. finalfsw/finalfsw.ino
Rocket `STATUS DESCENT`	Read monitor-only guidance diagnostics.	No steering outputs; linkage unverified. finalfsw/finalfsw.ino
CanSat `PWM25 <0..270deg | 500..2500us> / OFF`	CanSat MG996R GPIO25; OFF detaches/lowers signal.	Manual-anytime, either code mode. Cancels waiting hand-lift test; no mission/automatic-latch reset. OFF is not a position. cansatfswfinal/cansatfswfinal.ino
CanSat `STATUS PARACHUTE`	Read AT600, DEG90/US1167, mode, auto/test/output flags.	FEEDBACK=NONE; OUT is commanded pulse, not measured angle. cansatfswfinal/cansatfswfinal.ino
CanSat `PARACHUTE TEST / CANCEL`	Arm/cancel hand-lift detector:2m rise, >1m peak drop,5 fresh samples.	Bench flag1 required (current0). No explicit PAD gate; recovered unarmed/never-launched mission, good storage/reference and idle operations still required. CANCEL leaves PWM unchanged. cansatfswfinal/cansatfswfinal.ino
CanSat `STATUS RTC`	Read presence/validity/trust/advancing/oscillator/read status.	I2C ACK does not establish valid UTC. Set time after oscillator-stop. cansatfswfinal/cansatfswfinal.ino
CanSat `STATUS SPI`	Read SPI pins, cached flash JEDEC and initialized flags.	Does not prove journal health or RF delivery. cansatfswfinal/cansatfswfinal.ino
CanSat `STATUS PSMS`	Read PMS5003 status and valid atmospheric PM1/PM2.5/PM10.	ug/m3 only when valid; warmup/stale/frame faults are explicit. cansatfswfinal/cansatfswfinal.ino

COMMAND / DESTINATION	WHAT IT DOES	REQUIREMENTS AND EFFECTS
GCS only `TARGET ROCKET / TARGET CANSAT`	Select destination; Control dropdown does the same.	Both telemetry streams visible; one addressed command, no broadcast. Wait for pending request. workswithoutcmd copy.html
Ground `GROUND MAIN / BACKUP / STATUS`	Retune/query receiver only.	No vehicle command. Receiver USB: @RADIO MAIN/BACKUP/STATUS. workswithoutcmd copy.html; inspacelora/lora_rx/lora_rx.ino
CanSat USB only `ERASEFLASH CONFIRM / STATUS`	Full8MiB factory erase/verify/reboot or inspect progress.	Direct CanSat USB first60s only, recovered unarmed PAD/completed mission, telemetryOFF and idle. Deletes all mission/counter/calibration history. No remote erase authorized. cansatfswfinal/BootFlashErase.h; cansatfswfinal/cansatfswfinal.ino
Bench only `datafakecansat: ON / OFF / STATUS / BACKOFF ON / BACKOFF OFF / FLIGHT / RESETPAD CONFIRM / REBOOT`	Operate synthetic interference-test packets.	Not production commands; do not mix synthetic23:C stream with real CanSat. datafakecansat/datafakecansat.ino
Bench only `rocket_pcb_servo_test: 0..270 / u500..u2500 / c / l / r / s / f / d / a / ?`	Standalone Teensy servo test pin22(edit23).	s/f sweeps0..180; direct targets270. Startup moves90. Restore production after bench test. rocket_pcb_servo_test/rocket_pcb_servo_test.ino
Bench only `cansat_test_codes / I2C finder sketches`	Automatic diagnostics, no production command console.	Use sketch-specific wiring/readme then restore production. cansat_test_codes/README.md; cansat_i2c_finder/README.md; cansat_rtc_i2c_finder/README.md

05 / FAILURE GUIDE

A timeout is not a rejection

Before execution

GCS blocks, invalid bridge frames or RF losses may produce no flight ACK. Check selected vehicle, serial connection, receiver uplink output and matching channel. CanSat responding does not prove the rocket link.

After acceptance

OK may mean queued. Asynchronous commands can later return DONE/FAIL. Only matching valid ACKs confirm a request. A timeout means unknown outcome; no automatic retry is sent.

▶ **GCS will not send**

▶ **Receiver discards or waits**

▶ **Firmware silent validation drops**

▶ **Authentication / duplicates**

▶ **Real execution failures**

▶ **Async completion**

▶ **GCS rejects a reply or packet**

▶ **USB erase failures**

▶ **Physical delivery failures**

Source-linked negative diagnostic inventory

261 message/pattern locations in this source snapshot. Shared messages appear at multiple locations; printf placeholders remain patterns. Silent paths are explained above. See the cited source condition; no physical cause is inferred.

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `RADIO_SWITCH_PENDING`	finalfsw/finalfsw.ino line 1130 ▶ **Source condition**
REJECT `REBOOT_PENDING`	finalfsw/finalfsw.ino line 1133 ▶ **Source condition**
REJECT `PAD_RESET_PENDING`	finalfsw/finalfsw.ino line 1136 ▶ **Source condition**
REJECT `COUNTER_RESET_PENDING`	finalfsw/finalfsw.ino line 1140 ▶ **Source condition**
ERR `FLASH_READ_UNAVAILABLE;STORE=%u`	finalfsw/finalfsw.ino line 1172 ▶ **Source condition**
ERR `ARG`	finalfsw/finalfsw.ino line 1237 ▶ **Source condition**
ERR `ARG_UNSIGNED_MS`	finalfsw/finalfsw.ino line 1242 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `RANGE_%d_%d`	finalfsw/finalfsw.ino line 1244 ▶ **Source condition**
ERR `CAM_DISABLED`	finalfsw/finalfsw.ino line 1251 ▶ **Source condition**
ERR `ARG`	finalfsw/finalfsw.ino line 1252 ▶ **Source condition**
REJECT `CAM_BUSY`	finalfsw/finalfsw.ino line 1254 ▶ **Source condition**
REJECT `CAM_RECOVERY_OR_STORAGE_HOLD`	finalfsw/finalfsw.ino line 1256 ▶ **Source condition**
FAIL `CAM_STORAGE_FAILED`	finalfsw/finalfsw.ino line 1273 ▶ **Source condition**
FAIL `CAM_UART_FAILED_OR_DISABLED`	finalfsw/finalfsw.ino line 1281 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `ARG`	finalfsw/finalfsw.ino line 1291 ▶ **Source condition**
REJECT `CAM_RECORDING_REQUEST_ACTIVE`	finalfsw/finalfsw.ino line 1292 ▶ **Source condition**
FAIL `CAM_UART_FAILED_OR_DISABLED`	finalfsw/finalfsw.ino line 1294 ▶ **Source condition**
ERR `ARG_MAIN_OR_BACKUP`	finalfsw/finalfsw.ino line 1302 ▶ **Source condition**
ERR `RADIO_UNAVAILABLE`	finalfsw/finalfsw.ino line 1303 ▶ **Source condition**
REJECT `RADIO_BUSY`	finalfsw/finalfsw.ino line 1304 ▶ **Source condition**
ERR `ARG`	finalfsw/finalfsw.ino line 1315 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `RECOVERY_SCANNING`	finalfsw/finalfsw.ino line 1317 ▶ Source condition
REJECT `CAL_BUSY`	finalfsw/finalfsw.ino line 1318 ▶ Source condition
ERR `NO_BMP`	finalfsw/finalfsw.ino line 1319 ▶ Source condition
REJECT `IMU_STALE_OR_UNRELIABLE`	finalfsw/finalfsw.ino line 1320 ▶ Source condition
REJECT `KEEP_ROCKET_STILL`	finalfsw/finalfsw.ino line 1321 ▶ Source condition
REJECT `CAL_BARO_FIRST`	finalfsw/finalfsw.ino line 1322 ▶ Source condition
ERR `NO_RTC`	finalfsw/finalfsw.ino line 1336 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `RECOVERY_SCANNING`	finalfsw/finalfsw.ino line 1337 ▶ Source condition
ERR `ARG`	finalfsw/finalfsw.ino line 1339 ▶ Source condition
FAIL `RTC_VERIFY_FAILED`	finalfsw/finalfsw.ino line 1345 ▶ Source condition
ERR `NO_RTC`	finalfsw/finalfsw.ino line 1350 ▶ Source condition
REJECT `RECOVERY_SCANNING`	finalfsw/finalfsw.ino line 1356 ▶ Source condition
REJECT `CAM_BUSY`	finalfsw/finalfsw.ino line 1357 ▶ Source condition
REJECT `CAL_BUSY`	finalfsw/finalfsw.ino line 1358 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `USE_RESETPAD_CONFIRM`	finalfsw/finalfsw.ino line 1365 ▶ **Source condition**
REJECT `STORAGE_BUSY_OR_NOT_RECOVERABLE`	finalfsw/finalfsw.ino line 1366 ▶ **Source condition**
REJECT `CAL_OR_COUNTER_BUSY`	finalfsw/finalfsw.ino line 1367 ▶ **Source condition**
REJECT `STOP_RECORDING_BEFORE_PAD_RESET`	finalfsw/finalfsw.ino line 1368 ▶ **Source condition**
ERR `MISSION_ID_EXHAUSTED`	finalfsw/finalfsw.ino line 1369 ▶ **Source condition**
FAIL `PAD_RESET_NOT_QUEUED`	finalfsw/finalfsw.ino line 1372 ▶ **Source condition**
REJECT `STORAGE_NOT_RECOVERABLE`	finalfsw/finalfsw.ino line 1379 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `COUNTER_LEGACY_BASELINE_REQUIRED`	finalfsw/finalfsw.ino line 1381 ▶ Source condition
REJECT `COUNTER_EXHAUSTED`	finalfsw/finalfsw.ino line 1382 ▶ Source condition
REJECT `REVIEW_RECOVERY_THEN_ARM_CONFIRM`	finalfsw/finalfsw.ino line 1384 ▶ Source condition
ERR `ARG_CONFIRM_OR_NONE`	finalfsw/finalfsw.ino line 1386 ▶ Source condition
REJECT `STOP_RECORDING_BEFORE_ARM`	finalfsw/finalfsw.ino line 1388 ▶ Source condition
REJECT `IMU_STALE_OR_UNRELIABLE`	finalfsw/finalfsw.ino line 1389 ▶ Source condition
REJECT `KEEP_ROCKET_STILL`	finalfsw/finalfsw.ino line 1390 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `BARO_REFERENCE_NOT_READY`	finalfsw/finalfsw.ino line 1391 ▶ **Source condition**
REJECT `CAL_BUSY`	finalfsw/finalfsw.ino line 1392 ▶ **Source condition**
FAIL `ARM_NOT_PERSISTED`	finalfsw/finalfsw.ino line 1400 ▶ **Source condition**
ERR `USE_RESETCOUNT_CONFIRM`	finalfsw/finalfsw.ino line 1405 ▶ **Source condition**
REJECT `RESETCOUNT_NEEDS_TELEM_OFF_IDLE_KNOWN_HEALTHY_STORE`	finalfsw/finalfsw.ino line 1410 ▶ **Source condition**
FAIL `COUNTER_RESET_NOT_QUEUED`	finalfsw/finalfsw.ino line 1414 ▶ **Source condition**
REJECT `COUNTER_MIGRATION_NOT_AVAILABLE`	finalfsw/finalfsw.ino line 1422 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `ARG_LAST_USED_UPPER_BOUND`	finalfsw/finalfsw.ino line 1424 ▶ Source condition
ERR `COUNTER_RANGE`	finalfsw/finalfsw.ino line 1426 ▶ Source condition
ERR `UNKNOWN`	finalfsw/finalfsw.ino line 1433 ▶ Source condition
FAIL `CAL_STORAGE_FAILED_OLD_PARAMETERS_RETAINED`	finalfsw/finalfsw.ino line 1073 ▶ Source condition
FAIL `SENSOR_STALE_OR_UNRELIABLE`	finalfsw/finalfsw.ino line 1081 ▶ Source condition
FAIL `MOVED_OR_MISSION_ACTIVE`	finalfsw/finalfsw.ino line 1084 ▶ Source condition
FAIL `PRESSURE_UNSTABLE`	finalfsw/finalfsw.ino line 1090 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
FAIL `INSUFFICIENT_FRESH_BARO`	finalfsw/finalfsw.ino line 1102 ▶ Source condition
FAIL `INSUFFICIENT_FRESH_GYRO`	finalfsw/finalfsw.ino line 1103 ▶ Source condition
FAIL `INSUFFICIENT_FRESH_ACCEL`	finalfsw/finalfsw.ino line 1104 ▶ Source condition
FAIL `INVALID_CALIBRATION_OR_STORAGE_BUSY`	finalfsw/finalfsw.ino line 1109 ▶ Source condition
REJECT `AUTH_KEY`	finalfsw/finalfsw.ino line 1459 ▶ Source condition
REJECT `DELAYED_DUPLICATE`	finalfsw/finalfsw.ino line 1475 ▶ Source condition
ASYNC `CAM_NOT_SENT_STORAGE_OR_UART`	finalfsw/finalfsw.ino line 651 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ASYNC `PAD_RESET_STORAGE_FAILED_REBOOT_TO_RESOLVE`	finalfsw/finalfsw.ino line 805 ▶ Source condition
ASYNC `COUNTER_RESET_STORAGE_FAILED_REBOOT`	finalfsw/finalfsw.ino line 1747 ▶ Source condition
TRANSPORT/UI `FAILED, code`	finalfsw/finalfsw.ino line 850 ▶ Source condition
TRANSPORT/UI `# RTC calendar valid; UTC trust requires GNSS/TIME or journal`	finalfsw/finalfsw.ino line 925 ▶ Source condition
TRANSPORT/UI `# RTC invalid; waiting for GNSS/TIME, no compile-time adjustment`	finalfsw/finalfsw.ino line 925 ▶ Source condition
TRANSPORT/UI `# TX start failed, code`	finalfsw/finalfsw.ino line 977 ▶ Source condition
TRANSPORT/UI `# USB rejected: invalid command frame/checksum or wrong vehicle`	finalfsw/finalfsw.ino line 1440 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# CMD bad checksum`	finalfsw/finalfsw.ino line 1448 ▶ Source condition
TRANSPORT/UI `# USB rejected: invalid sequence`	finalfsw/finalfsw.ino line 1456 ▶ Source condition
TRANSPORT/UI `# CMD busy: previous radio ACK pending`	finalfsw/finalfsw.ino line 1465 ▶ Source condition
TRANSPORT/UI `# REBOOT: restarting after storage shutdown`	finalfsw/finalfsw.ino line 1719 ▶ Source condition
TRANSPORT/UI `# REBOOT: storage drain timed out; latest unsynced SD rows may be lost`	finalfsw/finalfsw.ino line 1719 ▶ Source condition
TRANSPORT/UI `# RESETCOUNT FAILED: storage fault; reboot for recovery`	finalfsw/finalfsw.ino line 1743 ▶ Source condition
TRANSPORT/UI `# USB partial command discarded: enable Newline or CRLF in Serial Monitor`	finalfsw/finalfsw.ino line 1752 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# USB rejected: invalid/oversized line`	finalfsw/finalfsw.ino line 1755 ▶ Source condition
TRANSPORT/UI `# USB rejected: use NAME,arg or NAME arg; one argument only`	finalfsw/finalfsw.ino line 1759 ▶ Source condition
ERR `ARG_ANGLE_0_270_OR_US_500_2500_OR_OFF`	cansatfswfinal/cansatfswfinal.ino line 1040 ▶ Source condition
ERR `PWM25_RANGE_ANGLE_0_270_OR_US_500_2500`	cansatfswfinal/cansatfswfinal.ino line 1043 ▶ Source condition
FAIL `PWM25_OUTPUT_FAILED;FEEDBACK=NONE`	cansatfswfinal/cansatfswfinal.ino line 1046 ▶ Source condition
REJECT `ESP32_ONLY`	cansatfswfinal/cansatfswfinal.ino line 1052 ▶ Source condition
REJECT `FLASH_MAINTENANCE_REBOOT_REQUIRED`	cansatfswfinal/cansatfswfinal.ino line 1059 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `RADIO_SWITCH_PENDING`	cansatfswfinal/cansatfswfinal.ino line 1063 ▶ **Source condition**
REJECT `REBOOT_PENDING`	cansatfswfinal/cansatfswfinal.ino line 1066 ▶ **Source condition**
REJECT `PAD_RESET_PENDING`	cansatfswfinal/cansatfswfinal.ino line 1069 ▶ **Source condition**
REJECT `COUNTER_RESET_PENDING`	cansatfswfinal/cansatfswfinal.ino line 1073 ▶ **Source condition**
ERR `FLASH_READ_UNAVAILABLE;STORE=%u`	cansatfswfinal/cansatfswfinal.ino line 1112 ▶ **Source condition**
ERR `ARG`	cansatfswfinal/cansatfswfinal.ino line 1187 ▶ **Source condition**
ERR `ARG_UNSIGNED_MS`	cansatfswfinal/cansatfswfinal.ino line 1192 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `RANGE_%d_%d`	cansatfswfinal/cansatfswfinal.ino line 1194 ▶ **Source condition**
ERR `ARG_MAIN_OR_BACKUP`	cansatfswfinal/cansatfswfinal.ino line 1202 ▶ **Source condition**
ERR `RADIO_UNAVAILABLE`	cansatfswfinal/cansatfswfinal.ino line 1203 ▶ **Source condition**
REJECT `RADIO_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1204 ▶ **Source condition**
ERR `ARG`	cansatfswfinal/cansatfswfinal.ino line 1215 ▶ **Source condition**
REJECT `RECOVERY_SCANNING`	cansatfswfinal/cansatfswfinal.ino line 1217 ▶ **Source condition**
REJECT `CAL_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1218 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERR `NO_BMP`	cansatfswfinal/cansatfswfinal.ino line 1219 ▶ Source condition
REJECT `IMU_STALE_OR_UNRELIABLE`	cansatfswfinal/cansatfswfinal.ino line 1220 ▶ Source condition
REJECT `KEEP_CanSat_STILL`	cansatfswfinal/cansatfswfinal.ino line 1221 ▶ Source condition
REJECT `CAL_BARO_FIRST`	cansatfswfinal/cansatfswfinal.ino line 1222 ▶ Source condition
ERR `NO_RTC`	cansatfswfinal/cansatfswfinal.ino line 1236 ▶ Source condition
REJECT `RECOVERY_SCANNING`	cansatfswfinal/cansatfswfinal.ino line 1237 ▶ Source condition
ERR `ARG`	cansatfswfinal/cansatfswfinal.ino line 1239 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
FAIL `RTC_VERIFY_FAILED`	cansatfswfinal/cansatfswfinal.ino line 1244 ▶ Source condition
ERR `NO_RTC`	cansatfswfinal/cansatfswfinal.ino line 1249 ▶ Source condition
ERR `ARG_TEST_OR_CANCEL`	cansatfswfinal/cansatfswfinal.ino line 1256 ▶ Source condition
REJECT `TEST_MODE_DISABLED_IN_CODE`	cansatfswfinal/cansatfswfinal.ino line 1257 ▶ Source condition
REJECT `TEST_REQUIRES_UNARMED_RECOVERED_MISSION`	cansatfswfinal/cansatfswfinal.ino line 1259 ▶ Source condition
REJECT `BARO_REFERENCE_NOT_READY`	cansatfswfinal/cansatfswfinal.ino line 1261 ▶ Source condition
REJECT `HANDLIFT_ALREADY_ARMED`	cansatfswfinal/cansatfswfinal.ino line 1263 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `ESP32_ONLY`	cansatfswfinal/cansatfswfinal.ino line 1268 ▶ **Source condition**
REJECT `RECOVERY_SCANNING`	cansatfswfinal/cansatfswfinal.ino line 1272 ▶ **Source condition**
REJECT `CAL_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1273 ▶ **Source condition**
ERR `USE_RESETPAD_CONFIRM`	cansatfswfinal/cansatfswfinal.ino line 1280 ▶ **Source condition**
REJECT `STORAGE_BUSY_OR_NOT_RECOVERABLE`	cansatfswfinal/cansatfswfinal.ino line 1281 ▶ **Source condition**
REJECT `CAL_OR_COUNTER_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1282 ▶ **Source condition**
ERR `MISSION_ID_EXHAUSTED`	cansatfswfinal/cansatfswfinal.ino line 1283 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
FAIL `PAD_RESET_NOT_QUEUED`	cansatfswfinal/cansatfswfinal.ino line 1286 ▶ **Source condition**
REJECT `BENCH_BUILD_ARM_DISABLED`	cansatfswfinal/cansatfswfinal.ino line 1292 ▶ **Source condition**
REJECT `STORAGE_NOT_RECOVERABLE`	cansatfswfinal/cansatfswfinal.ino line 1294 ▶ **Source condition**
REJECT `COUNTER_LEGACY_BASELINE_REQUIRED`	cansatfswfinal/cansatfswfinal.ino line 1296 ▶ **Source condition**
REJECT `COUNTER_EXHAUSTED`	cansatfswfinal/cansatfswfinal.ino line 1297 ▶ **Source condition**
REJECT `REVIEW_RECOVERY_THEN_ARM_CONFIRM`	cansatfswfinal/cansatfswfinal.ino line 1299 ▶ **Source condition**
ERR `ARG_CONFIRM_OR_NONE`	cansatfswfinal/cansatfswfinal.ino line 1301 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `IMU_STALE_OR_UNRELIABLE`	cansatfswfinal/cansatfswfinal.ino line 1303 ▶ **Source condition**
REJECT `KEEP_CanSat_STILL`	cansatfswfinal/cansatfswfinal.ino line 1304 ▶ **Source condition**
REJECT `BARO_REFERENCE_NOT_READY`	cansatfswfinal/cansatfswfinal.ino line 1305 ▶ **Source condition**
REJECT `CAL_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1306 ▶ **Source condition**
FAIL `ARM_NOT_PERSISTED`	cansatfswfinal/cansatfswfinal.ino line 1313 ▶ **Source condition**
ERR `USE_RESETCOUNT_CONFIRM`	cansatfswfinal/cansatfswfinal.ino line 1318 ▶ **Source condition**
REJECT `RESETCOUNT_NEEDS_TELEM_OFF_IDLE_KNOWN_HEALTHY_STORE`	cansatfswfinal/cansatfswfinal.ino line 1323 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
FAIL `COUNTER_RESET_NOT_QUEUED`	cansatfswfinal/cansatfswfinal.ino line 1327 ▶ **Source condition**
REJECT `COUNTER_MIGRATION_NOT_AVAILABLE`	cansatfswfinal/cansatfswfinal.ino line 1335 ▶ **Source condition**
ERR `ARG_LAST_USED_UPPER_BOUND`	cansatfswfinal/cansatfswfinal.ino line 1337 ▶ **Source condition**
ERR `COUNTER_RANGE`	cansatfswfinal/cansatfswfinal.ino line 1339 ▶ **Source condition**
ERR `UNKNOWN`	cansatfswfinal/cansatfswfinal.ino line 1346 ▶ **Source condition**
FAIL `CAL_STORAGE_FAILED_OLD_PARAMETERS_RETAINED`	cansatfswfinal/cansatfswfinal.ino line 989 ▶ **Source condition**
FAIL `SENSOR_STALE_OR_UNRELIABLE`	cansatfswfinal/cansatfswfinal.ino line 997 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
FAIL `MOVED_OR_MISSION_ACTIVE`	cansatfswfinal/cansatfswfinal.ino line 1000 ▶ Source condition
FAIL `PRESSURE_UNSTABLE`	cansatfswfinal/cansatfswfinal.ino line 1006 ▶ Source condition
FAIL `INSUFFICIENT_FRESH_BARO`	cansatfswfinal/cansatfswfinal.ino line 1018 ▶ Source condition
FAIL `INSUFFICIENT_FRESH_GYRO`	cansatfswfinal/cansatfswfinal.ino line 1019 ▶ Source condition
FAIL `INSUFFICIENT_FRESH_ACCEL`	cansatfswfinal/cansatfswfinal.ino line 1020 ▶ Source condition
FAIL `INVALID_CALIBRATION_OR_STORAGE_BUSY`	cansatfswfinal/cansatfswfinal.ino line 1025 ▶ Source condition
REJECT `AUTH_KEY`	cansatfswfinal/cansatfswfinal.ino line 1372 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
REJECT `DELAYED_DUPLICATE`	cansatfswfinal/cansatfswfinal.ino line 1388 ▶ **Source condition**
ASYNC `PAD_RESET_STORAGE_FAILED_REBOOT_TO_RESOLVE`	cansatfswfinal/cansatfswfinal.ino line 722 ▶ **Source condition**
ASYNC `COUNTER_RESET_STORAGE_FAILED_REBOOT`	cansatfswfinal/cansatfswfinal.ino line 1632 ▶ **Source condition**
TRANSPORT/UI `# SX1262 pins: SCK=%d MISO=%d MOSI=%d CS=%d RESET=%d BUSY=%d\n`	cansatfswfinal/cansatfswfinal.ino line 764 ▶ **Source condition**
TRANSPORT/UI `FAILED, code`	cansatfswfinal/cansatfswfinal.ino line 771 ▶ **Source condition**
TRANSPORT/UI `# SX1262 RF-switch configuration failed: %d\n`	cansatfswfinal/cansatfswfinal.ino line 785 ▶ **Source condition**
TRANSPORT/UI `# IMU initialization failed at 0x69`	cansatfswfinal/cansatfswfinal.ino line 830 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# RTC calendar valid; UTC trust requires TIME or journal`	cansatfswfinal/cansatfswfinal.ino line 863 ▶ Source condition
TRANSPORT/UI `# RTC invalid; waiting for TIME, no compile-time adjustment`	cansatfswfinal/cansatfswfinal.ino line 863 ▶ Source condition
TRANSPORT/UI `# TX start failed, code`	cansatfswfinal/cansatfswfinal.ino line 893 ▶ Source condition
TRANSPORT/UI `# USB rejected: invalid command frame/checksum or wrong vehicle`	cansatfswfinal/cansatfswfinal.ino line 1353 ▶ Source condition
TRANSPORT/UI `# CMD bad checksum`	cansatfswfinal/cansatfswfinal.ino line 1361 ▶ Source condition
TRANSPORT/UI `# USB rejected: invalid sequence`	cansatfswfinal/cansatfswfinal.ino line 1369 ▶ Source condition
TRANSPORT/UI `# CMD busy: previous radio ACK pending`	cansatfswfinal/cansatfswfinal.ino line 1378 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# REBOOT: restarting after storage shutdown`	cansatfswfinal/cansatfswfinal.ino line 1600 ▶ Source condition
TRANSPORT/UI `# REBOOT: storage drain timed out; latest unsynced SD rows may be lost`	cansatfswfinal/cansatfswfinal.ino line 1600 ▶ Source condition
TRANSPORT/UI `# RESETCOUNT FAILED: storage fault; reboot for recovery`	cansatfswfinal/cansatfswfinal.ino line 1628 ▶ Source condition
TRANSPORT/UI `# USB partial command discarded: enable Newline or CRLF in Serial Monitor`	cansatfswfinal/cansatfswfinal.ino line 1637 ▶ Source condition
TRANSPORT/UI `# USB rejected: invalid/oversized line`	cansatfswfinal/cansatfswfinal.ino line 1640 ▶ Source condition
TRANSPORT/UI `# USB rejected: use NAME,arg or NAME arg; one argument only`	cansatfswfinal/cansatfswfinal.ino line 1644 ▶ Source condition
TRANSPORT/UI `# ERASEFLASH REJECT:`	cansatfswfinal/cansatfswfinal.ino line 1662 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# HANDLIFT PWM FAILED; NO RETRY`	cansatfswfinal/cansatfswfinal.ino line 1869 ▶ Source condition
TRANSPORT/UI `Receiver command blocked: connect serial, stop simulation, wait for pending command.`	workswithoutcmd copy.html line 456 ▶ Source condition
TRANSPORT/UI `Invalid command envelope/team/key/checksum; nothing sent.`	workswithoutcmd copy.html line 467 ▶ Source condition
TRANSPORT/UI `FAILED at ground bridge:`	workswithoutcmd copy.html line 521 ▶ Source condition
TRANSPORT/UI `. Physical execution unconfirmed.`	workswithoutcmd copy.html line 521 ▶ Source condition
TRANSPORT/UI `FAILED: connection lost; execution unknown. No retry sent.`	workswithoutcmd copy.html line 534 ▶ Source condition
TRANSPORT/UI `Rocket-only command blocked for CanSat; nothing sent.`	workswithoutcmd copy.html line 542 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `FAILED: serial write; execution unknown. No retry sent.`	workswithoutcmd copy.html line 577 ▶ Source condition
TRANSPORT/UI `Invalid ACK ignored`	workswithoutcmd copy.html line 597 ▶ Source condition
TRANSPORT/UI `# RADIO invalid receiver argument: MAIN BACKUP STATUS`	inspacelora/lora_rx/lora_rx.ino line 97 ▶ Source condition
TRANSPORT/UI `# RX rearm failed:`	inspacelora/lora_rx/lora_rx.ino line 112 ▶ Source condition
TRANSPORT/UI `# UPLINK partial line timed out`	inspacelora/lora_rx/lora_rx.ino line 120 ▶ Source condition
TRANSPORT/UI `# UPLINK invalid or oversized line`	inspacelora/lora_rx/lora_rx.ino line 136 ▶ Source condition
TRANSPORT/UI `# RX rearm failed:`	inspacelora/lora_rx/lora_rx.ino line 167 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# Radio init failed, code:`	inspacelora/lora_rx/lora_rx.ino line 222 ▶ Source condition
TRANSPORT/UI `# startReceive failed, code:`	inspacelora/lora_rx/lora_rx.ino line 234 ▶ Source condition
TRANSPORT/UI `Read failed, code:`	inspacelora/lora_rx/lora_rx.ino line 307 ▶ Source condition
REJECT `PWM%u_UNCONFIGURED`	finalfsw/ServoControl.h line 190 ▶ Source condition
ERR `PWM%u_RANGE_%u_%u_US`	finalfsw/ServoControl.h line 192 ▶ Source condition
ERASE `ALREADY_ATTEMPTED_REBOOT_REQUIRED`	cansatfswfinal/BootFlashErase.h line 18 ▶ Source condition
ERASE `BOOT_WINDOW_CLOSED_REBOOT_REQUIRED`	cansatfswfinal/BootFlashErase.h line 19 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
ERASE `NEED_RECOVERED_UNARMED_PAD_OR_COMPLETED_MISSION_AND_IDLE_COMMANDS`	cansatfswfinal/BootFlashErase.h line 20 ▶ **Source condition**
ERASE `CHIP_BUSY_PROTECTED_UNAVAILABLE_OR_WRONG_ID`	cansatfswfinal/BootFlashErase.h line 21 ▶ **Source condition**
ERASE `TIMEOUT_REBOOT_REQUIRED`	cansatfswfinal/BootFlashErase.h line 27 ▶ **Source condition**
ERASE `VERIFY_READ_FAILED_REBOOT_REQUIRED`	cansatfswfinal/BootFlashErase.h line 37 ▶ **Source condition**
ERASE `NOT_FULLY_ERASED_REBOOT_REQUIRED`	cansatfswfinal/BootFlashErase.h line 40 ▶ **Source condition**
TRANSPORT/UI `# RX failed %d\n`	datafakecansat/datafakecansat.ino line 41 ▶ **Source condition**
TRANSPORT/UI `# TX start failed %d\n`	datafakecansat/datafakecansat.ino line 47 ▶ **Source condition**

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
TRANSPORT/UI `# NVS failed: numbered telemetry blocked`	datafakecansat/datafakecansat.ino line 118 ▶ Source condition
TRANSPORT/UI `# Radio init failed %d; check lora_tx wiring/module\n`	datafakecansat/datafakecansat.ino line 169 ▶ Source condition
TRANSPORT/UI `# TX completion timeout (no delivery/execution confirmation)`	datafakecansat/datafakecansat.ino line 183 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 650 ▶ Source condition
SOURCE CONDITION `RECOVERED_SEQ_OUTCOME_UNKNOWN`	finalfsw/finalfsw.ino line 760 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 804 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1073 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1081 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1084 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1090 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1102 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1103 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1104 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1109 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1273 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1281 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1294 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1345 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1372 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1400 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1414 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `COUNTER_STORAGE_FAILED`	finalfsw/finalfsw.ino line 1429 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1430 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1583 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1646 ▶ Source condition
SOURCE CONDITION `FAIL`	finalfsw/finalfsw.ino line 1746 ▶ Source condition
SOURCE CONDITION `NOT_FOUND`	cansatfswfinal/cansatfswfinal.ino line 498 ▶ Source condition
SOURCE CONDITION `RECOVERED_SEQ_OUTCOME_UNKNOWN`	cansatfswfinal/cansatfswfinal.ino line 675 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 721 ▶ Source condition
SOURCE CONDITION `NOT_FOUND`	cansatfswfinal/cansatfswfinal.ino line 857 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 989 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 997 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1000 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1006 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1018 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1019 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1020 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1025 ▶ Source condition
SOURCE CONDITION `PWM25_OFF;NO_POSITION_FEEDBACK`	cansatfswfinal/cansatfswfinal.ino line 1037 ▶ Source condition
SOURCE CONDITION `PWM25_OFF_FAILED;OUTPUT_UNKNOWN`	cansatfswfinal/cansatfswfinal.ino line 1037 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1038 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1046 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1244 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1286 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1313 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1327 ▶ Source condition
SOURCE CONDITION `COUNTER_STORAGE_FAILED`	cansatfswfinal/cansatfswfinal.ino line 1342 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1343 ▶ Source condition
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1528 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `FAIL`	cansatfswfinal/cansatfswfinal.ino line 1631 ▶ Source condition
SOURCE CONDITION `FAILED`	cansatfswfinal/cansatfswfinal.ino line 1734 ▶ Source condition
SOURCE CONDITION `ARG_SPIN_REVERSE_STOP_SAFE_TEST90_OR_ANGLE_OR_US`	finalfsw/ServoControl.h line 188 ▶ Source condition
SOURCE CONDITION `ARG_SAFE_TEST90_OR_ANGLE_0_270_OR_US_500_2500`	finalfsw/ServoControl.h line 188 ▶ Source condition
SOURCE CONDITION `DELAYED_DUPLICATE`	datafakecansat/datafakecansat.ino line 69 ▶ Source condition
SOURCE CONDITION `ARG`	datafakecansat/datafakecansat.ino line 76 ▶ Source condition
SOURCE CONDITION `USE_RESETPAD_CONFIRM`	datafakecansat/datafakecansat.ino line 82 ▶ Source condition

MESSAGE / RESULT FAMILY	SOURCE EVIDENCE
SOURCE CONDITION `ARG`	datafakecansat/datafakecansat.ino line 87 ▶ **Source condition**
SOURCE CONDITION `RANGE_400_10000`	datafakecansat/datafakecansat.ino line 90 ▶ **Source condition**

Data: command catalog · failure catalog · source SHA256 snapshot.

06 / PROJECT SUMMARY

Two vehicles. One GCS. Separate state.

Browser GCS ↔ USB ground receiver ↔ **LoRa** ↔ Rocket23:R / CanSat23:C

SUBSYSTEM	ROCKET	CANSAT
MCU	Teensy 4.1	ESP32-WROOM
SX1262 SPI	MOSI11/MISO12/SCK13; CS2/reset6/BUSY4; RXEN1/TXEN0	MOSI23/MISO19/SCK18; CS5/reset12/BUSY13; polled RX
NOR	W25Q64JV8MiB, CS10	W25Q64JV8MiB, CS4, EF4017
I2C sensors	BMP581/BNO08x: Wire2 SDA25/SCL24. DS3231: Wire18/19. INA219: Wire1 SDA17/SCL16.	Wire SDA21/SCL22,100 kHz. INA219 0x40; BMP581 0x47 (fallback0x46); DS3231 0x68; MPU6500-compatible 0x69.
Outputs / UART	PWM4 pin22/PWM5 pin23. RunCam Serial5 TX20/RX21,115200.	PWM25 GPIO25. PMS5003 Serial2 RX26/TX27,9600.
GNSS / SD	Existing GNSS and built-in SD CSV logging.	No GPS, no SD. GNSS fields stay zero; valid RTC time or00:00:00.

Automatic actions and timing

- Wire states:0PAD,1POWERED_ASCENT,2COAST,3APOGEE,4PARACHUTE_DESCENT,5LANDED. Existing sensor-fusion/freshness/sustained transitions remain. Mission elapsed starts on detected launch.
- Rocket PWM5 stows0 on known unarmed PAD and commands90 after durable APOGEE→DESCENT intent. PWM4 automatic linkage stays unconfigured; manual full0..270 range remains available.
- CanSat stow0/release90 (500/1167us). Flight flag0. DESCENT, peak>600m, negative speed and5 fresh samples with fused+barometric height≤600m relative pad are required. No arbitrary delay; a never-above600m flight has no invented automatic fallback.
- Automatic output waits for durable intent. Uncertain prior actions are not blindly replayed after reset. No ACK proves servo travel/parachute deployment.
- Rocket locally sends one camera stop60s after confirmed landing if recording was requested. GCS need not stay open. Both start/stop use experimental0x01 toggle packet CC0101E7; no physical recording feedback.
- Telemetry stays16-field checksummed CSV, separate vehicle counters/epochs and command sequences. Bridge preserves raw machine-readable packets/ACKs. CanSat IMU charts use CanSat time arrays.

Known limits: launch/flight thresholds, nominal270deg servo mapping and release mechanics need physical validation. RTC must be valid/advancing/trusted for reset continuity. Shared-channel RF mitigation is not collision-proof. PMS warmup/stale/frame faults are explicit; numerical PM concentrations are not added to the fixed recovery-record layout. Successful compilation is not a hardware test.

07 / EVIDENCE & SHARING

Verification record

PASS: ten focused software/documentation suites and installed firmware/object checks. No physical hardware validation; existing compiler/runtime environment warnings remain.

CHECK	OBSERVED RESULT	EVIDENCE
`node tests/command-phase-policy.test.cjs`	PASS: actual CAL/REBOOT bodies allow nonPAD; stale-data and busy failures retained.	log
`node tests/remote-counter.test.cjs`	PASS: both flight handlers/durable epoch/timeout/reply cache and actual selected GCS frames/ACKs.	log
`node tests/telemetry-check.cjs`	PASS:30s OFF has zero routine TX; ACK survives; ON resumes; explicit OFF allowed in recovery/flight.	log
`node tests/usb-command-path.cjs`	PASS: real GCS frames/actual dispatch/reply, authentication, duplicate suppression, asynchronous completion.	log
`node tests/targets-check.cjs`	PASS: target isolation/shared GCS/source pin/storage guards.	log

CHECK	OBSERVED RESULT	EVIDENCE
`node tests/cansat-parachute.test.cjs`	PASS: actual hand-lift handler retains execution requirements and bench mode.	log
`node tests/servo.test.cjs`	PASS: existing servo console/snippet/ACK routing and documented handler inventory.	log
`node tests/gcs-cansat-plots.test.cjs`	PASS: real CanSat packet/parser/Plotly data arrays use correct time axis, including interleaving and invalid gaps.	log
`node tests/cansat-erase.test.cjs`	PASS: original USB-only erase/protection/full verification/no-retry behavior remains.	log
`powershell -File cansatfswfinal/check.ps1`	PASS: ESP32 flight0/flash-disabled/bench1 objects; MPU dependency; actual C++ journal/600m/one-shot assertions.	log
`powershell -File verification_results/descent-build.ps1`	PASS: rocket/legacy CanSat/Servo/PCB bench ARM objects, journal/parser assertions, receiver syntax. No linked firmware image.	log
`node tests/handoff-doc.test.cjs`	PASS:41 command families, source-linked reasons, source hashes/links/anchors, offline JS and search filters; no hardware API.	log

Actual-source policy tests substitute IO; installed compiler checks generate objects, not uploaded/physically validated systems. Existing warnings remain in logs.

Share the source

ZIP complete finalfsw/, cansatfswfinal/, receiver code, common GCS HTML, this folder and required documents/pinouts. Include every local header/subfolder. Exclude nested inspace/ workspace copy, build artifacts, editor settings and private API keys. ZIP does not apply .gitignore. Open this index.html offline; Print/Save PDF needs no development tools.

Short commands · Protocol · Feature history · CanSat deployment · Sharing/setup